THE ANH NGUYEN

Machine Learning, AI and Data Engineer

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PROFESSIONAL SUMMARY

Master of Artificial Intelligence graduate from UTS (Computer Vision sub-major, GPA 6.67/7, Dean's List 2026) with an engineering background in control and automation. I build machine learning systems from the data pipeline to deployment and measure them on the target hardware: YOLO detection in a mobile browser, INT8 detection on a Raspberry Pi 5 and a reinforcement learning policy on a Jetson Nano robot car. Hands-on with PyTorch, computer vision, reinforcement learning, MLOps (ClearML, Optuna, GitHub Actions) and backend services (FastAPI, Docker, PostgreSQL). Available now.

TECHNICAL SKILLS

Programming: Python, SQL, TypeScript
Machine Learning / Deep Learning: PyTorch, scikit-learn, Transformers, CNNs, LSTM/GRU, XGBoost, SHAP
Computer Vision: YOLO, RF-DETR, RT-DETR, Faster R-CNN, SAM 2, OpenCV, object detection, multi-object tracking
Reinforcement Learning: PPO, DQN, SAC, Stable-Baselines3, sim-to-real transfer, Donkey Car simulator, Isaac Sim
Edge Deployment: ONNX, ONNX Runtime and ONNX Runtime Web, INT8 quantisation, Jetson Nano, Raspberry Pi 5
MLOps: ClearML, Optuna, GitHub Actions, AWS SageMaker Studio, experiment tracking, model versioning
Backend / Deployment: FastAPI, Docker, PostgreSQL, Streamlit, REST APIs
Tools: Git, GitHub, Linux, Jupyter, pytest, Vitest

EDUCATION

Master of Artificial Intelligence | University of Technology Sydney, Sydney, Australia *Feb 2025 – Nov 2026*
Sub-major: Computer Vision **GPA:** 6.67/7 **Award:** Dean's List 2026, Faculty of Engineering and IT
Relevant coursework: Image Processing and Pattern Recognition · CNNs and Deep Learning · Reinforcement Learning · Advanced Data Analytics

Bachelor of Engineering in Control and Automation (Advanced Program) *Sep 2020 – Sep 2024*
Hanoi University of Science and Technology (HUST), Hanoi, Vietnam
Major: Automation Control Engineering **Sub-major:** Artificial Intelligence **GPA:** 3.65/4

EXPERIENCE

AI / Computer Vision Intern | FPT Software, Hanoi, Vietnam *Dec 2025 – Feb 2026*

- Studied the Segment Anything Model 2 (SAM 2) architecture, focusing on its prompt encoder, mask decoder, memory encoder and temporal memory-attention components for image and video segmentation.
- Reimplemented key components of the SAM 2 architecture in Python to understand model structure, tensor flow, module interactions and segmentation behaviour.
- Used LandingAI to segment and prepare input images for downstream SAM 2 experiments, exploring how preprocessing and region extraction affected segmentation workflows.
- Ran proof-of-concept experiments on image segmentation tasks and analysed model behaviour across different prompts and visual inputs.

PROJECTS

VoiceEye: Real-Time AI Vision Assistant *Apr – May 2026*
Computer Vision · Edge AI · MLOps github.com/BushMar12/VoiceEye | Live: voiceeye.pages.dev

- Curated a **48-class object-detection dataset** for blind-navigation scenarios by merging COCO 2017, Open Images v7 and Mapillary Vistas, adding six safety-critical classes, with automated validation and stratified train/validation splits.
- Trained and tuned a **YOLO26n** detector to **0.50 mAP@50 and 0.35 mAP@50–95** with Optuna Bayesian hyperparameter optimisation.
- Built a reproducible **ClearML MLOps pipeline** for data preparation, training, evaluation, export and quality-gated model promotion, integrated with GitHub Actions training and deployment workflows.
- Deployed the model for **fully on-device browser inference** with ONNX Runtime Web at about **10 FPS on mobile** (throttled for battery), cut the runtime bundle from 24 MB to about 12 MB, and fixed browser memory crashes through buffer reuse and tensor lifecycle management.
- Implemented a real-time perception pipeline in TypeScript with multi-object tracking, monocular distance estimation, hazard-based attention prioritisation and latency/FPS monitoring, covered by **144 automated unit tests**.

JetRacer RL: Sim-to-Real Autonomous Driving (Team Project, UTS) *Sep – Oct 2026*

Reinforcement Learning · Robotics · Edge AI Private repository, code walkthrough on request

- Main code contributor in a team training a camera-only Waveshare JetRacer (Jetson Nano) in the Donkey Car simulator and driving it on a real lab track. Owned the **obstacle-stop policy, simulator training pipeline and on-car runtime**.
- Narrowed the sim-to-real gap by matching the simulator to the real camera (160° fisheye lens model, camera pose, colour normalisation) and randomising shading, glare, mat creases and steering response during training.
- Trained a **DQN STOP/GO policy** (Stable-Baselines3) beside the team's frozen PPO steering policy after a PPO version failed to stop reliably. Rendered obstacle approach clips in Isaac Sim (1.6 m to contact in 1 cm steps) and exported both policies as one ONNX model.
- Diagnosed a simulator-only shortcut and removed it by adding real lab backgrounds to training frames. Selected checkpoints on held-out real lab frames, raising **AUROC from 0.52 to 0.89**.
- In the lab test the car **stopped for 14 of 14 objects** in its path, running a **20 Hz control loop with 6.9 ms median inference** on the Jetson Nano. Also built a browser dashboard for calibration and driving, a log replay tool and an AWS SageMaker Studio setup for teammates without a GPU.

Intelligent Vehicle Damage Assessment System (Team Project, UTS) *Apr 2026*

Computer Vision · Object Detection · MLOps · Generative AI github.com/BushMar12/Intelligent-Vehicle-Damage-Assessment

- Team project building an end-to-end system that detects and localises six vehicle-damage categories, assesses severity, estimates repair costs in AUD and generates assessment reports from images and videos. I maintain the public repository.
- Augmented the **CarDD dataset from 1,971 to 5,913 training images**, standardised inputs to 640×640 and benchmarked six detection architectures, including YOLO, RT-DETR, RF-DETR and Faster R-CNN.
- RF-DETR-m reached **76.87% mAP@0.5**. The team deployed **YOLO26m** for its size and speed: 72.53% mAP@0.5, 74.12% F1, 42 MB and about 23 ms GPU inference latency.
- Delivered ClearML workflows for dataset versioning, experiment tracking, model promotion and inference monitoring, and a Dockerised FastAPI and PostgreSQL service with a Qwen 2.5 assistant served through Ollama.

Smart Parking Management System Using YOLOv8 *2024*

Bachelor's Graduation Thesis, HUST *Computer Vision · Edge AI · Embedded Systems*

- Designed and developed a real-time smart parking prototype using YOLOv8n, combining licence-plate detection at the entrance with parking-space occupancy detection.
- Built a hardware setup around a Raspberry Pi 5 with two cameras: one at the entrance for licence plates and one monitoring parking bays for free spaces.
- Deployed and optimised the pipeline on the Raspberry Pi 5 with **INT8 quantisation**, reaching about **30 FPS with 90%+ detection accuracy**.
- Sent availability to an LCD at the parking entrance and a web monitoring interface, and applied pre- and post-processing to keep detection robust under changing light.

Multimodal Deep Learning for Stock Return Prediction (In Progress) *2026*

Deep Learning · Time Series · Data Ingestion github.com/BushMar12/Tesla-stock-price-prediction-using-Multimodal

- Building an end-to-end multimodal pipeline for next-day TSLA return prediction that fuses 60-day OHLCV and technical-indicator sequences with Alpha Vantage news sentiment.
- Designed a **1.67M-parameter two-stream architecture**: a Transformer price encoder and a multi-kernel 1D CNN sentiment encoder joined by multi-head cross-modal attention, benchmarked against LSTM, GRU, Transformer and XGBoost baselines with leakage-safe chronological splits.
- Running a 3-configuration × 5-seed ablation study with SHAP-based explainability, plus a 7-tab Streamlit dashboard and rate-limit-aware market data ingestion. Final results pending.

ADDITIONAL INFORMATION

Languages: Vietnamese (Native) | English (Professional Working Proficiency)

Research Interests: Computer Vision, Multimodal AI, Vision-Language-Action Models, Embodied AI, Reinforcement Learning

Strengths: Technical communication · Analytical problem-solving · Team collaboration · Self-directed learning